

PROBLEM

Visualization Recommendation (VisRec) algorithms frequently embed visualization expertise, yet the origin and authenticity of this expert knowledge often remain ambiguous. It is essential to recognize that preferred visualizations and their practical utility can vary significantly across different domains and problem areas. Consequently, specific domains may require unique sets of visualizations that differ from those used in other domains. By considering these diverse requirements during the design process, a more practical and effective VisRec strategy can be achieved, enhancing the usability and relevance for the intended audience.

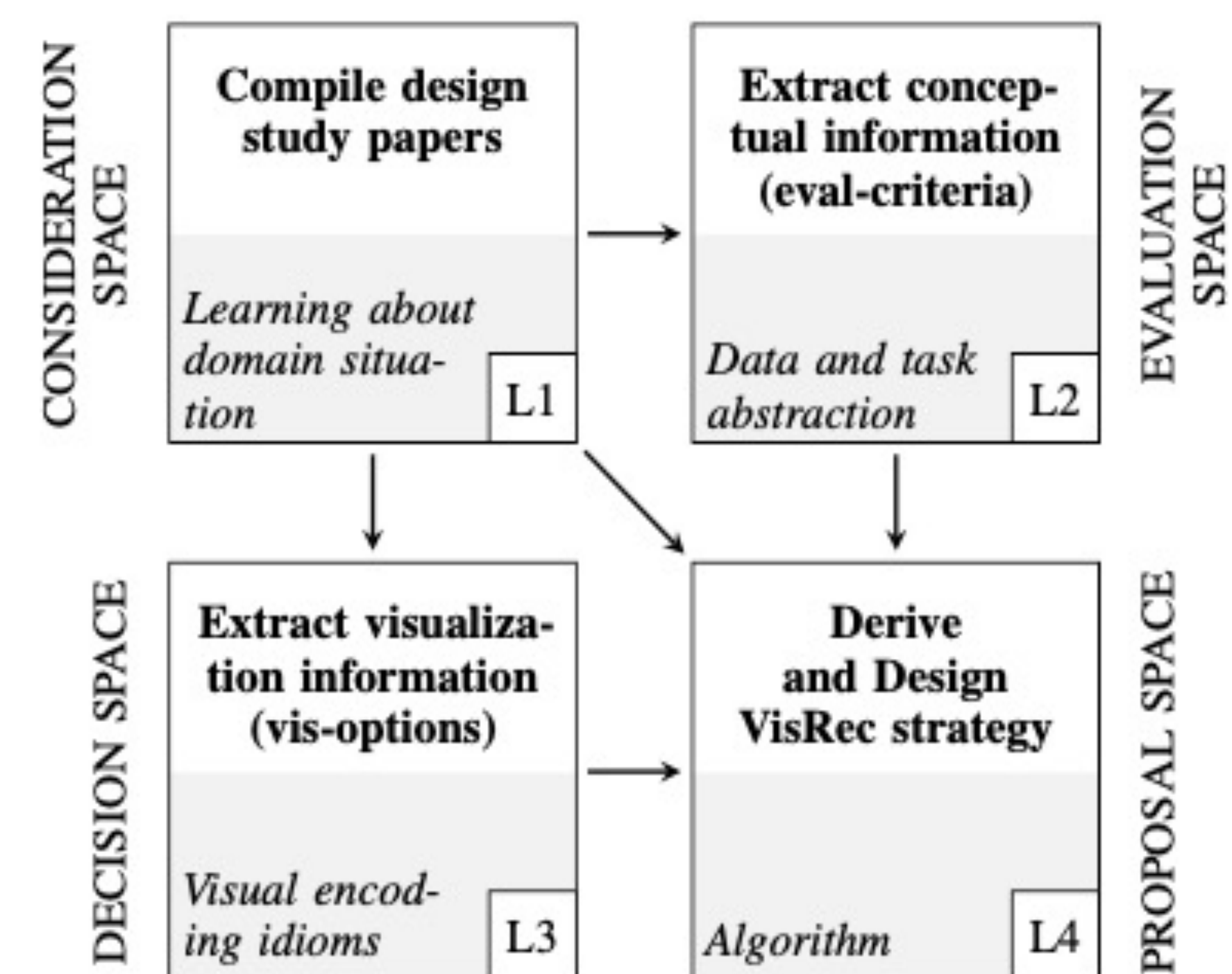
We introduce the concept of a **meta design study**, wherein visualization knowledge is systematically derived from design study papers in a certain domain or area, ensuring that a resulting VisRec strategy is grounded in the peer-reviewed knowledge of visualization experts.

THEORY

The conventional framework for design studies follows the steps of the nested model of visualization design (learning about domain situation [L1]; data- and task-abstraction [L2]; derive visual encoding idioms [L3]; algorithmic considerations [L4]) in a top-down manner [3]. From a particular viewpoint, the visualization researcher conducting such a design study serves as a VisRec oracle, transforming a wide *Consideration Space* into a narrow *Proposal Space* of visualization options [5]. This perspective aligns with the concept that a full Design Space actually consists of a *Decision Space* (in our context, the visualization options to choose from) and an *Evaluation Space* (criteria for selecting appropriate options from the Decision Space) [2].

Based on these observations, a *meta design study* provides all components to formulate a *VisRec strategy*. A visualization researcher **compiles design study papers** for the problem area under investigation. Afterward, they **extract conceptual information** and **visualization information** from those compilations which can further be utilized to **derive and design a VisRec strategy**.

CONCEPT



PRACTICAL EXAMPLE

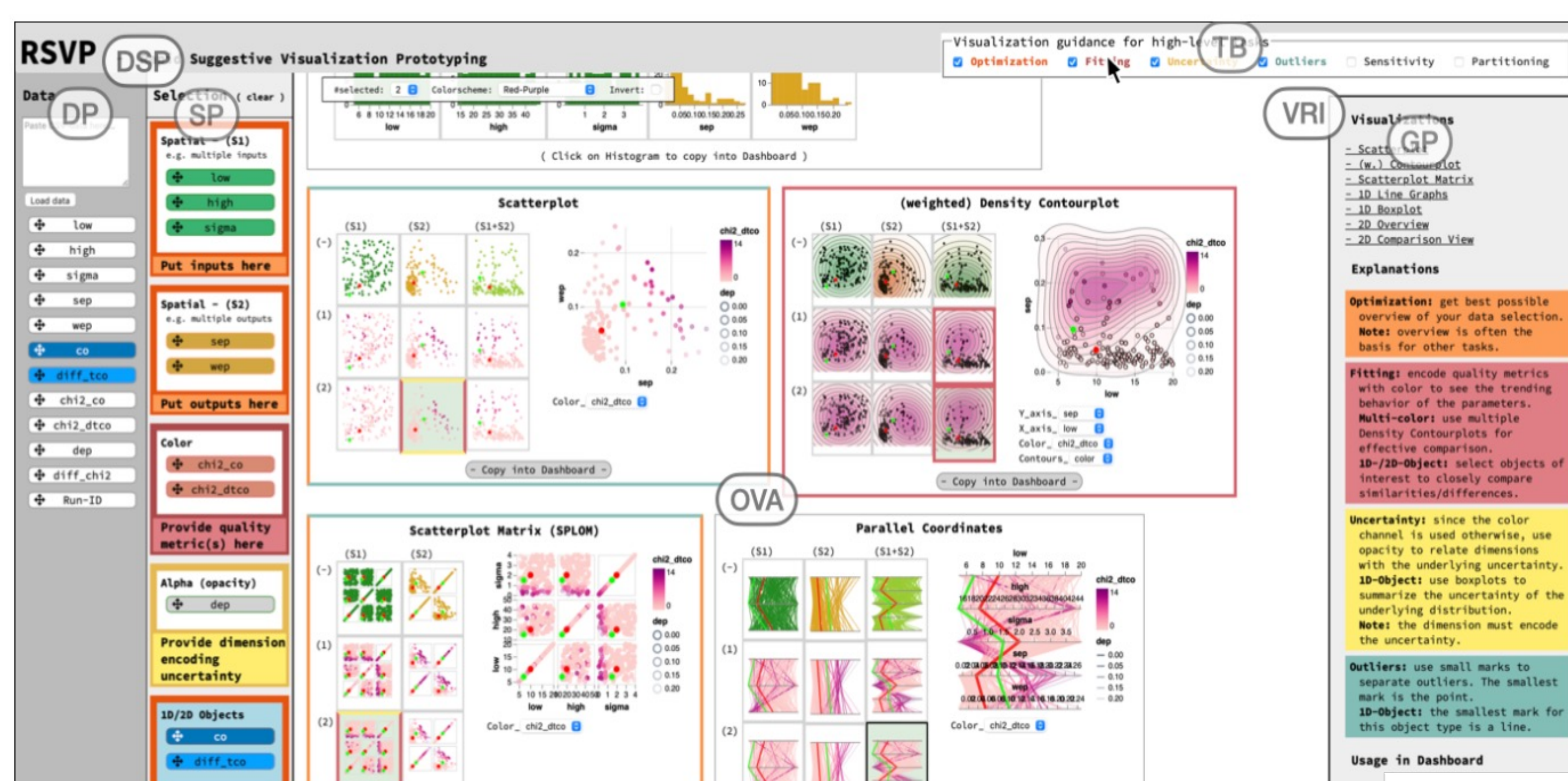
EVAL CRITERIA L2

Data	Type	
Sampling	Multi-dimensional/ Multi-variate	(MDMV)
	Complex Object	(CO)
	Regular	(reg)
	Stochastic	(stoch)

Tasks		
Optimization	Find best parameter setting	T_{opt}
Fitting	Where does actual data occur	T_{fit}
Uncertainty	How reliable is the output	T_{unc}
Outliers	Find odd or special outputs	T_{out}
Sensitivity	Regions with high / low impact	T_{sens}
Partitioning	Discern different model behavior	T_{part}

PAPER COMPILATION L1

We collected and analyzed **45 design study papers** involving **Visual Parameter Space Analysis (VPSA)** [4].



RSVP, the **Rapid Suggestive Visualization Prototyping System** [1], is derived from the insights of a meta design study over the body of knowledge on *Visual Parameter Space Analysis (VPSA)*. It implements the design space and a multi-view VisRec strategy from the meta design study in a non-obtrusive manner. The linked video presents an overview of the system; details and findings of the meta design study are highlighted in the surrounding information blocks.

VIS OPTIONS L3

MDMV	CO
(SP) Scatterplot	
(wDCP) Density Contourplot	
(SPLOM) Scatterplot Matrix	
(rSPLOM) reduced SPLOM	
(PSc) Point Scales	
(PC) Parallel Coordinates	
(Hist) Histograms	
(1D-Line) 1D - Linegraph	
(1D-Box) 1D - Boxplot	
(1D-Hist) 1D - Cumulative Hist	
(2D-Grid) 2D - Gridview	
(2D-Jux) 2D - Juxtapos. views	
(2D-Sup) 2D - Superpos. views	

VISREC STRATEGY L4

App	Hist	SP	SPLOM	rSPLOM	PSc	PC	wDCP	CO	Task	Sampl
#1	S:5			S:5	S:5				T_{opt}	reg
#2	S:4			S:3+C+B	S:4			1D-Box	T_{fit}	stoch
#3	S:6	S:8 (1)				S:8		2D-Grid	T_{unc}	stoch
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

Excerpt showcasing a coding table with enhanced visual elements. The headers in white and blue indicate visualization options, light gray headers correspond to evaluation criteria. These criteria serve as filters for table rows, resulting in a refined display of applications and the corresponding visualizations that address the chosen criteria.

Task	Strategy	MDMV	CO
T_{opt}	Overview	<u>Spatial</u> expressivity	1D-Line, 2D-Grid
T_{fit}	Affiliation	Overview + <u>Color</u>	2D-Jux
T_{unc}	Attenuation	Overview + <u>Brightness</u>	1D-Box
T_{out}	Separation	<u>Point</u> -based (0D-mark)	(1D-Line)
T_{sens}	Con-/Divergence	<u>Line</u> -based (1D-mark)	1D-Hist, 2D-Sup
T_{part}	Summarization	<u>Area</u> -based (2D-mark)	2D-Grid (-)

High-level overview of the derived rules for the task-oriented part of the VisRec strategy for VPSA applications. We found that each task has a general solving strategy independent of the underlying data type. We further found that for MDMV types, tasks are typically associated with a distinct mark or channel. T_{opt}

Dims#	In-reg	In-stoch; Out
(1)	(PSc+Hist)	(PSc)
2	wDCP+Hist	SP
3	SPLOM+wDCP+Hist	SPLOM
4	PC+wDCP+Hist	(r)SPLOM
5	PC+wDCP+Hist	rSPLOM
6	PC+Hist	rSPLOM
7-9	PC	PC
10+	(PC)	PSc

Guidelines for *spatial expressivity* to maximize data representation while minimizing occlusion and visual clutter. Rules are based on encoded dimensions, the sampling strategy, and if inputs or outputs are visualized.

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- The QR code was generated with *QR-Code Generator & QR Creator* for iOS.

REFERENCES

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